

23CSE452
BUSINESS ANALYTICS

User Preferences for Recommendation Systems: A Survey-Based Case Study of YouTube

Case Study Report

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1. Problem Statement and Objectives

1.1 Business Problem

Recommendation systems are used by digital platforms to help users find content without manually searching for every item. However, user interaction with recommendations varies. A recommendation may be ignored because it is irrelevant, repetitive, too long, or poorly presented, while other recommendations may lead users to continue watching or discover new creators.

This case study uses YouTube as a case study for recommendation systems and examines recommendation interaction from the user's perspective. Instead of studying only clicks, the study collects user-reported viewing behaviour, recommendation reliance, reasons for accepting or rejecting recommendations, perceived repetition, discovery behaviour, preferred recommendation strategies, satisfaction, and willingness to watch more when recommendations match interests.

The main analytical target is recommendation watch percentage, defined in the survey as the percentage of videos watched that come from YouTube recommendations. The study then uses regression models to examine whether behavioural and preference-related variables can predict this measure.

1.2 Objectives

- Identify behavioural and preference-related factors associated with interaction with recommended content.
- Examine how recommendation reliance relates to satisfaction, viewing time, repetition, skip behaviour, and recommendation-surface usage.
- Predict recommendation watch percentage using regression models.
- Translate the observed findings into practical recommendations for recommendation-system design.

1.3 Scope

The study is user-centred rather than an implementation of a production recommendation algorithm. YouTube is used as the case study, while the findings are discussed in terms of recommendation systems more generally. The study does not claim to reproduce YouTube's internal ranking algorithm.

1.4 Expected Business Use

The analysis can help identify where users experience recommendation friction and which recommendation behaviours are associated with greater reliance. These findings can be used as inputs when considering ranking, diversity, discovery, and user-control decisions.

2. Data Collection and Dataset Description

2.1 Primary Data Collection

The primary dataset was collected through a Google Forms questionnaire. A total of 115 verified responses were obtained. No personal identity variables were used in the analytical dataset. The questionnaire was designed around YouTube usage and recommendation interaction.

- The survey contained 18 original fields covering the following areas:
- YouTube usage frequency and weekly viewing hours.
- Percentage of videos watched from recommendations.
- Reasons for clicking or ignoring recommended videos.

- Recommendation surfaces used, such as Home Feed, Shorts and Subscriptions.
- Actions taken after ignoring a recommendation and the number of videos skipped.
- Perceived recommendation problems, including repetition, irrelevance and clickbait.
- Creator discovery through recommendations and Shorts.
- Types of recommendations users want more of and what they think YouTube should prioritize.
- Overall satisfaction and likelihood of watching more if recommendations better match interests.

2.2 Main Variables

Variable	Type	Role in Analysis
recommendation_watch_pct	Numeric	Main target: recommendation reliance
weekly_hours	Numeric	Viewing intensity
satisfaction_score	1–10	Perceived system satisfaction
skip_count_numeric	Discrete	Recommendation friction
repetition_frequency	Ordinal	Perceived repetition
creator_discovery_frequency	Ordinal	New-creator discovery
surfaces_count	Count	Breadth of recommendation surfaces
watch_more_likelihood	Ordinal	Potential engagement response

2.3 Dataset Scaling

Because the assignment required at least 5,000 records, the 115 real survey responses were retained as the empirical seed dataset and expanded to 10,000 synthetic records. An empirical Gaussian copula with correlated kernel resampling was used: observed rows were bootstrap-sampled, correlations among numeric variables were preserved, correlated perturbations were added, and values were mapped back to the empirical distributions. Further implementation detail is provided in the accompanying analysis notebook.

Validation showed close agreement between the seed and synthetic data. KS p-values were 1.0000 for weekly hours, 0.9268 for recommendation watch percentage, 1.0000 for satisfaction and 1.0000 for skip friction. The detailed validation tables and correlation comparisons are provided in the accompanying analysis notebook.

3. Data Preparation and Exploratory Analysis

3.1 Data Preparation

The 18 raw survey columns were cleaned by removing the timestamp, renaming fields to concise snake_case names, standardising labels and converting the main numeric fields. Excel had incorrectly formatted some skip ranges as dates; these were restored to 1-5, 6-10 and 11-20, with More than 20 represented as >20. Further implementation detail is provided in the accompanying analysis notebook.

Skip ranges were converted to skip_count_numeric using midpoints of 3, 8, 15 and 25. This provided a numeric proxy for skip friction while retaining the original categories for grouped analysis. Missing Shorts responses were represented as Never / NA and category labels were standardised. Further cleaning details are provided in the accompanying analysis notebook.

Four multi-option questions (click_reasons, ignore_reasons, surfaces_used and desired_recommendations) were converted into three types of features: selection counts, behavioural archetypes and binary thematic flags. Counts captured the number of selections, archetypes grouped combinations into broader patterns, and flags

retained individual themes. The original multi-value columns were then removed. Further mapping details are provided in the accompanying analysis notebook.

Engineered Features

The engineered features preserved both the breadth of each respondent's selections and the main behavioural theme. For modelling, categorical variables were one-hot encoded, while ordered variables such as repetition frequency, creator discovery frequency and watch-more likelihood were converted to numeric scales. This produced a 60-column feature matrix. Further implementation details are provided in the accompanying analysis notebook.

Source response	Engineered feature(s)	How it was derived	Purpose in analysis
click_reasons	click_reasons_count	Count the selected click-reason options.	Measures how many criteria are considered before clicking.
click_reasons	click_archetype	Classify selected themes as Topic_Driven, Creator_Driven, Packaging_Driven, Content_Creator_Blend or Packaging_Influenced.	Represents the dominant click motivation.
click_reasons	click_topic_flag; click_creator_flag; click_packaging_flag	Set each flag to 1 when the corresponding topic, creator or packaging theme is present; otherwise 0.	Retains specific click-driver information after compression.
ignore_reasons	ignore_reasons_count	Count the selected reasons for ignoring a recommendation.	Measures the breadth of rejection/friction reasons.
ignore_reasons	ignore_archetype	Classify responses as Relevance_Friction, Packaging_Friction, Duration_Friction or Compound_Friction.	Summarises the main type of rejection friction.
ignore_reasons	ignore_relevance_flag; ignore_packaging_flag; ignore_duration_flag	Set binary indicators according to whether each rejection theme appears.	Separates relevance, packaging and duration effects.
surfaces_used	surfaces_count	Count the selected recommendation surfaces.	Measures breadth of recommendation-surface use.
surfaces_used	surface_archetype	Classify navigation as Home_Exclusive, Search_Oriented, Shorts_Primary, Home_Hybrid, Omni_Surface or Subscription_Primary.	Represents the respondent's dominant navigation pattern.
surfaces_used	surface_home_flag; surface_shorts_flag; surface_subscriptions_flag	Set a binary flag for regular use of Home, Shorts and Subscriptions.	Retains individual surface-adoption signals.
desired_recommendations	desired_count	Count the recommendation types selected as desirable.	Measures breadth of recommendation expectations.
desired_recommendations	discovery_orientation	Classify preferences as Exploitation_Focused, Exploration_Focused or Balanced_Discovery.	Represents preference for familiar-content reinforcement versus exploration.
desired_recommendations	desired_exploitation_flag; desired_exploration_flag	Set binary indicators for familiar-content and novel-	Retains the two main discovery axes.

		content preference themes.	
skip_count_range	skip_count_numeric	Map 1-5→3, 6-10→8, 11-20→15 and >20→25.	Provides a numeric proxy for skip friction.

3.2 Empirical Survey Findings

The 115 real responses provide the baseline for the analysis. More than 62% of respondents reported using YouTube daily or multiple times a day. Across all respondents, recommendations accounted for an average of 41.9% of reported video watch percentage.

Viewing intensity varied by usage frequency. Users who reported using YouTube multiple times a day averaged 9.58 hours per week and 52.85% recommendation watch percentage. Daily users averaged 5.44 hours and 45.87% recommendation watch percentage. Users using YouTube several times a week averaged 3.91 hours and 25.91% recommendation watch percentage.

3.3 Main Recommendation Friction

The most common reported issues were irrelevant videos (33.0%) and repetitive videos (32.2%). Too much clickbait accounted for 13.9%, while too many Shorts accounted for 12.2%. These responses indicate that recommendation quality was not viewed only as a relevance problem; repetition and presentation were also reported as sources of friction.

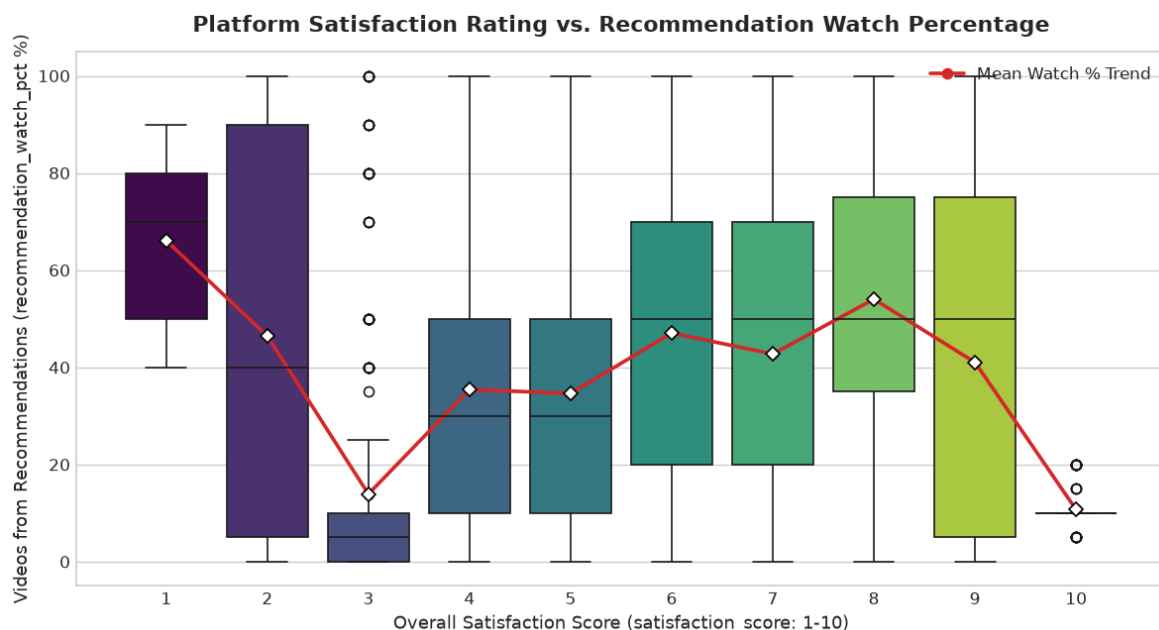
More than 70% of respondents skip between 1 and 5 recommended videos before choosing one. Users reporting clickbait or irrelevance as important friction points were more likely to report larger skip counts. Further breakdowns of skip behaviour are provided in the accompanying analysis notebook.

3.4 Exploratory Relationships

The exploratory analysis uses the 10,000-record synthetic cohort to examine relationships between recommendation watch percentage and selected behavioural variables. These analyses are intended to identify patterns rather than establish causal relationships.

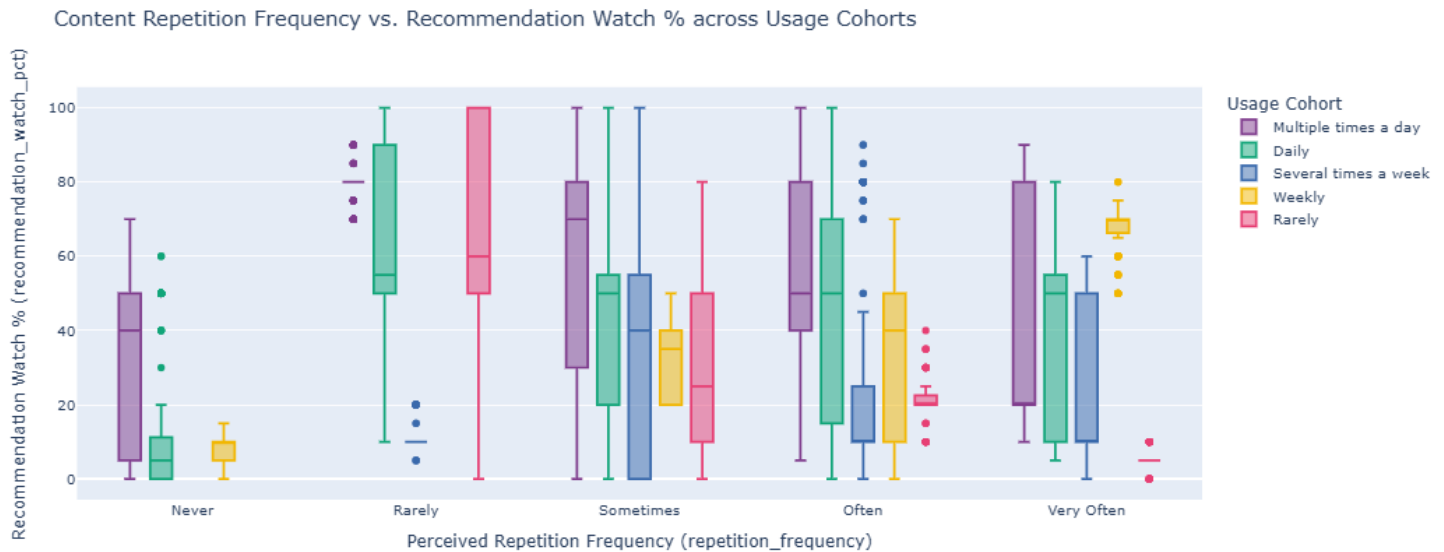
Satisfaction and Recommendation Reliance

Users with satisfaction scores from 1 to 4 had a mean recommendation watch percentage of 38.8%. Users with satisfaction scores of 8 or higher had a mean of 48.7%. This is consistent with a positive association between satisfaction and recommendation reliance.



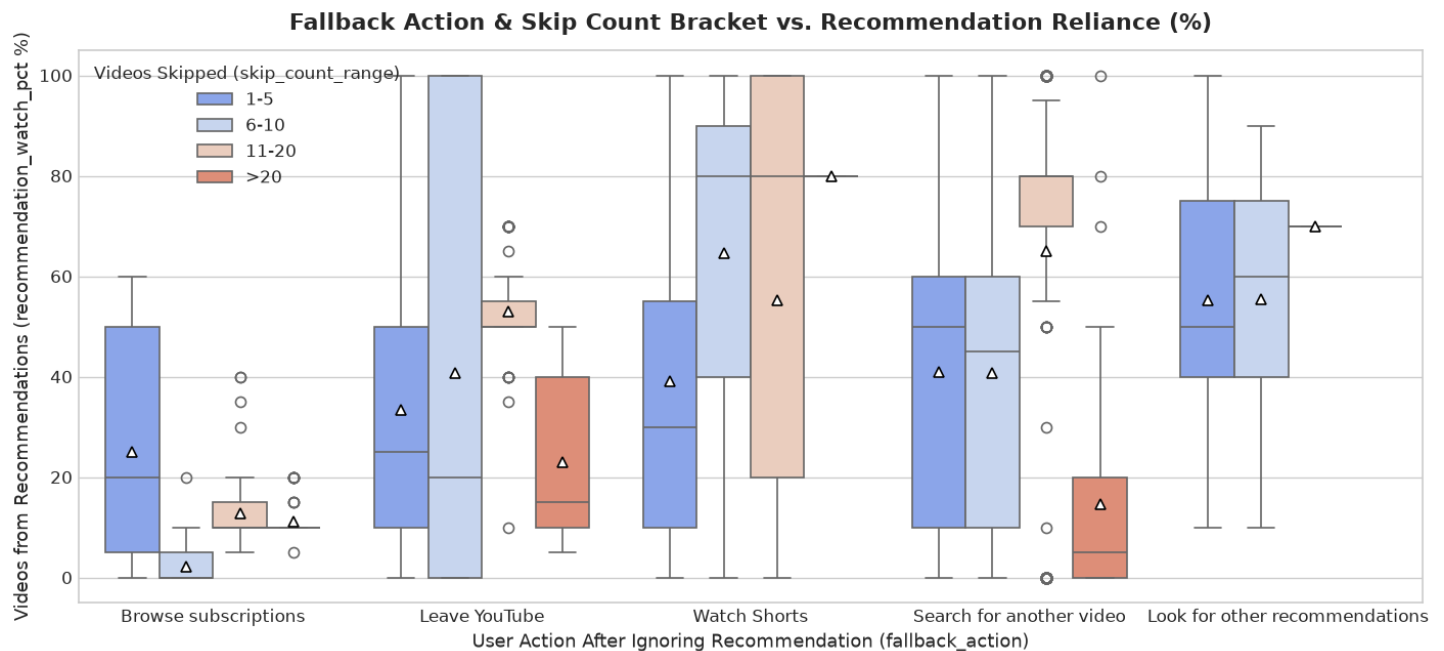
Repetition and Recommendation Reliance

Among daily users, the median recommendation watch percentage was 50% when repetition was reported as Rarely and 40% when repetition was reported as Very Often. For users who accessed YouTube multiple times a day, the corresponding medians were 60% and 50%. The pattern is consistent with lower recommendation reliance when repetition is reported more often.



Skip Friction and Fallback Behaviour

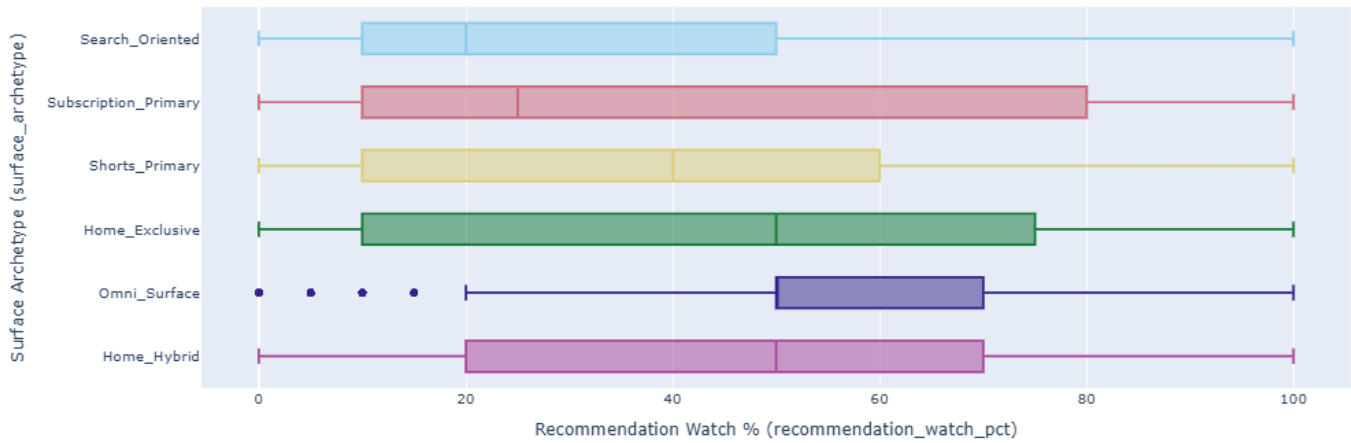
Users who selected 'Look for other recommendations' after ignoring a recommendation had the highest mean recommendation watch percentage at 55.3%. Users who selected 'Leave YouTube' after skipping more than 20 videos had a mean of 22.9% and a median of 15%. This indicates that high skip friction combined with leaving the platform is associated with low recommendation reliance.



Recommendation Surface Usage

Users classified as Omni_Surface users, who regularly use Home, Shorts and Subscriptions, had a mean recommendation watch percentage of 53.6%. Search_Oriented users had the lowest mean at 22.9%. This suggests that broader use of recommendation surfaces is associated with greater reliance on recommendations.

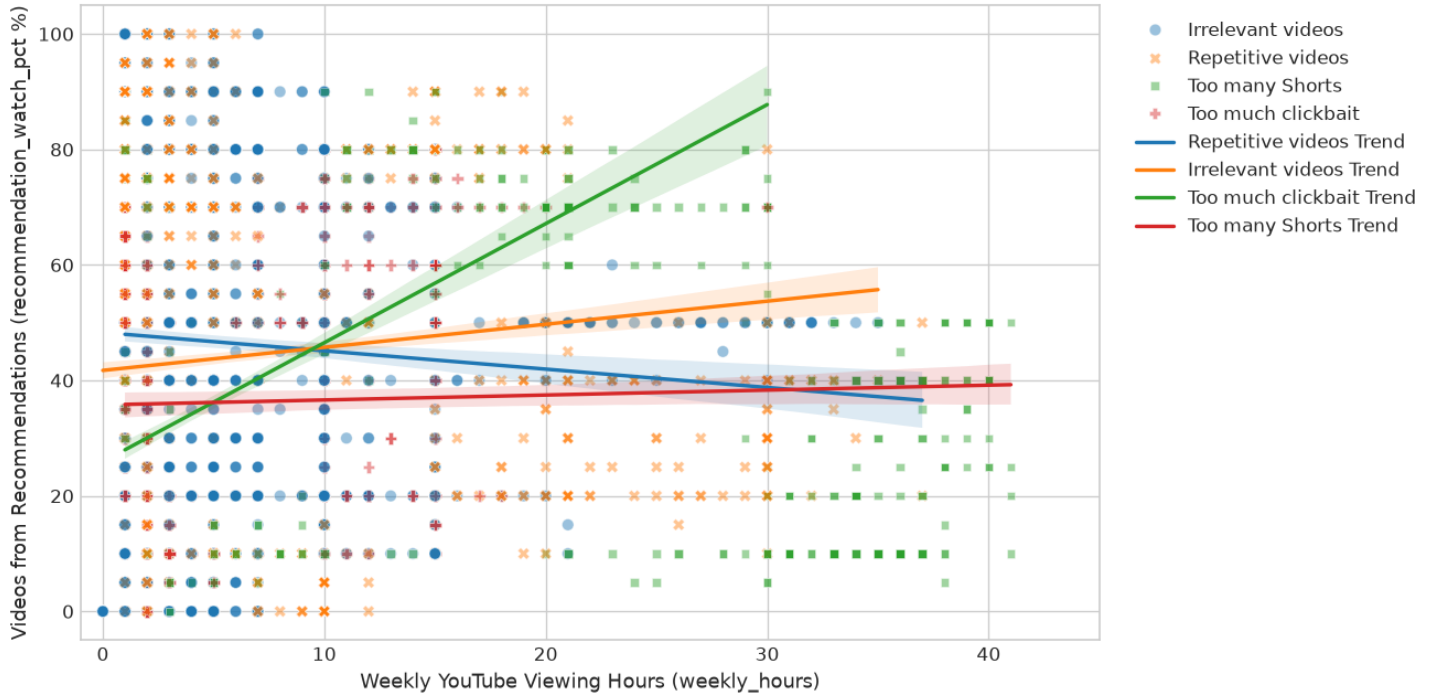
Recommendation Surface Archetype vs. Recommendation Watch Percentage



Viewing Hours and Recommendation Reliance

Weekly viewing hours had a positive correlation with recommendation watch percentage ($r = +0.110$). Users spending 15 or more hours per week had a mean recommendation watch percentage of 53.4% and a median of 50.0%. Among the major friction categories, repetitive videos and too much clickbait showed flatter trends, indicating that friction can limit recommendation reliance as viewing hours increase.

Weekly Viewing Hours vs. Recommendation Watch % by Primary Algorithmic Issue



4. Analytics Method and Implementation

4.1 Analytical Approach

The primary predictive task is regression. The target variable is `recommendation_watch_pct`, a value from 0 to 100 representing the percentage of viewed videos coming from recommendations. Two models were selected to provide a linear baseline and a non-linear ensemble comparison.

4.2 Ridge Regression

Ridge Regression was used as the baseline model. It provides a simple linear relationship between the engineered predictors and recommendation watch percentage. The model was implemented in a pipeline with standardisation and Ridge regularisation.

4.3 Random Forest Regressor

Random Forest Regression was used to capture non-linear relationships and interactions between behavioural variables. The model used 100 trees, a maximum depth of 12 and a minimum leaf size of 5. Further implementation detail is provided in the accompanying analysis notebook.

4.4 Feature Preparation

The engineered features were combined with the original single-response variables to form the modelling matrix. Ordinal variables—repetition_frequency, creator_discovery_frequency and watch_more_likelihood—were mapped to ordered numeric scales. Categorical variables were one-hot encoded with the first category removed. Numeric predictors included weekly_hours, satisfaction_score, skip_count_numeric, click_reasons_count, ignore_reasons_count, surfaces_count and desired_count. The engineered binary flags and archetype variables were also retained after encoding. The resulting feature matrix contained 60 predictors. The data was split into 80% training and 20% testing data using random_state=42, and five-fold K-fold cross-validation was applied. The full preprocessing-to-model pipeline and encoded feature matrix are provided in the accompanying analysis notebook.

The resulting feature matrix contained 60 predictors. The data was split into 80% training and 20% testing data using a fixed random state of 42. Five-fold cross-validation was applied to the training data.

4.5 Evaluation Metrics

- R^2 : proportion of target variance explained by the model.
- Adjusted R^2 : R^2 adjusted for the number of predictors.
- RMSE: average prediction error with larger errors receiving more weight.
- MAE: average absolute prediction error in percentage points.
- Five-fold cross-validation R^2 : checks whether performance is consistent across training folds.

4.6 Model Performance

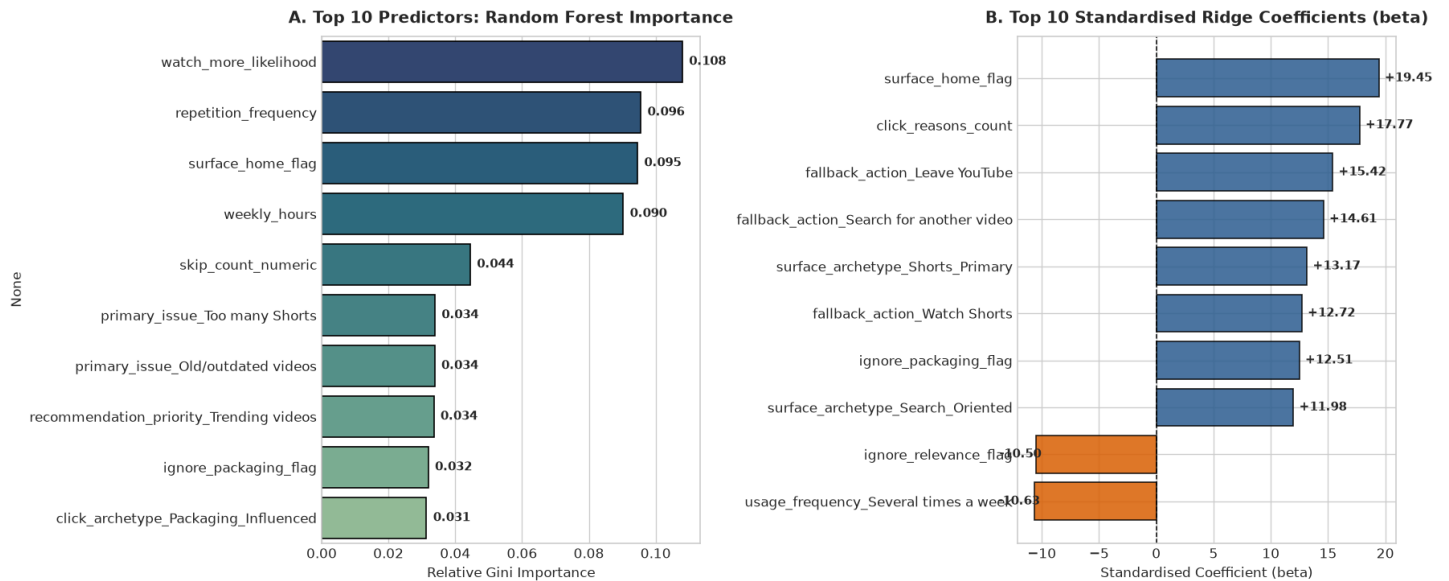
Model	Train R^2	Test R^2	Adj. R^2	RMSE	MAE
Ridge Regression	0.5445	0.5263	0.5116	20.73	16.49
Random Forest	0.9696	0.9659	0.9649	5.56	3.39

Random Forest produced a test R^2 of 0.9659, with RMSE 5.56 percentage points and MAE 3.39 percentage points. Its five-fold cross-validation R^2 was 0.9656 ± 0.0021 . Ridge produced a test R^2 of 0.5263, RMSE 20.73 and MAE 16.49, with cross-validation R^2 of 0.5368 ± 0.0069 .

4.7 Feature Importance and Correlation Analysis

The Random Forest model identified watch_more_likelihood (0.108), repetition_frequency (0.096), surface_home_flag (0.095) and weekly_hours (0.090) as its four highest-importance predictors. These importance values describe predictive contribution within the fitted model and should not be interpreted as

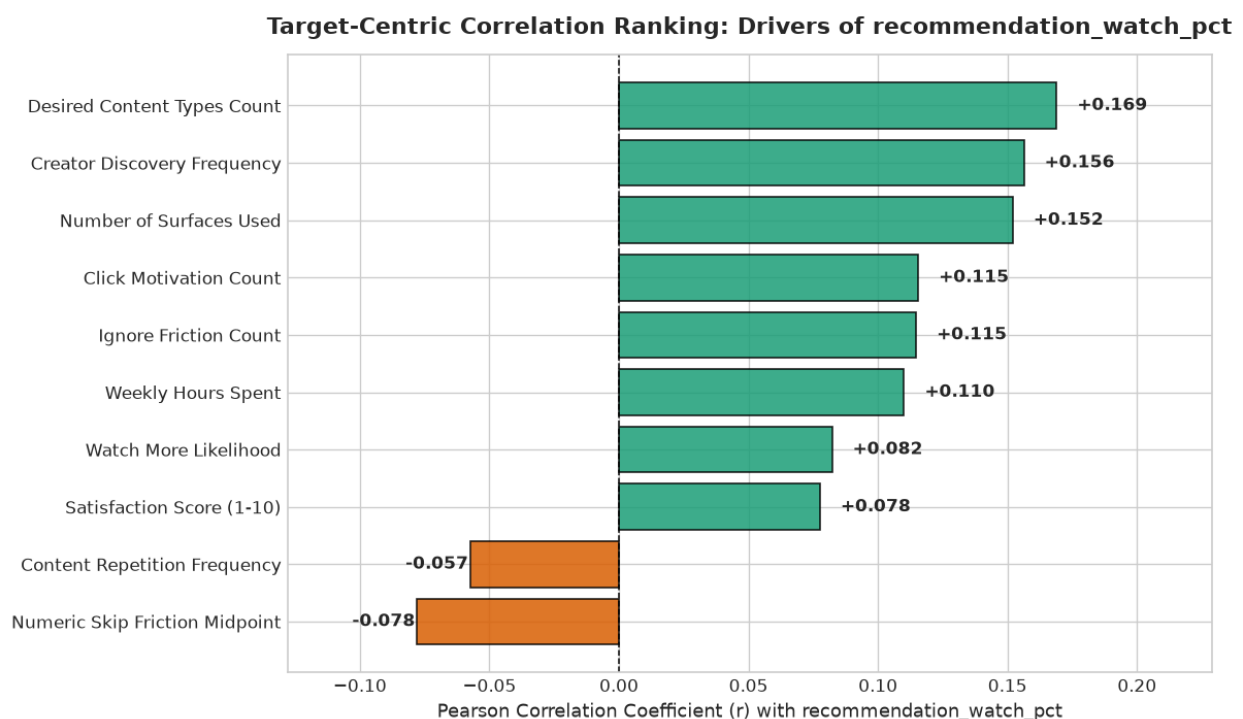
causal effects. The corresponding feature-importance visualization is provided in the accompanying analysis notebook.



The Ridge coefficients provide a different interpretation of the same modelling matrix. The Home Feed indicator, `surface_home_flag`, had the strongest positive standardised coefficient among the reported variables (+19.45), while the 'Several times a week' usage category had the strongest negative coefficient among the displayed variables (-10.63). These coefficients describe the fitted linear association after standardisation and regularisation. The complete coefficient output is provided in the accompanying analysis notebook.

4.8 Target-Centric Correlation Results

Pearson correlations with recommendation watch percentage were generally modest. The strongest positive correlations were `surfaces_count` ($r = +0.152$), `weekly_hours` ($r = +0.110$) and `satisfaction_score` ($r = +0.078$). The strongest negative correlations were `skip_count_numeric` ($r = -0.078$) and `repetition_frequency_num` ($r = -0.057$). These are pairwise associations and do not imply causality. The complete correlation ranking and underlying calculations are provided in the accompanying analysis notebook.



4.9 Interpretation of Model Results

The difference between the two models indicates that recommendation reliance is not well represented by a simple linear relationship alone in this dataset. The Random Forest captured non-linear combinations of viewing behaviour, surface usage, repetition and stated engagement intention that were not captured as well by Ridge.

However, the high Random Forest score should be interpreted with the synthetic-data limitation in mind. The model was trained and tested on synthetic records generated from only 115 real responses. Therefore, the reported predictive performance should not be treated as equivalent to performance on a large independent sample of real users.

5. Comparison with State-of-the-Art Methods

Recent recommender-system studies provide context for the present case study. The comparison focuses on method, dataset, evaluation approach, findings, and how each study differs from the present user-survey analysis. Detailed study-level comparison is kept in the report table, while implementation and analysis details for the present case study are provided in the accompanying analysis notebook.

Published Study / Year	Dataset	Method	Evaluation	Key Result	Comparison with This Work
Wu et al. (2024)	YouTube production traffic	Learned Ranking Function; slate optimisation; multi-objective constrained optimisation	Live experiments / A-B testing	Reports improvements in a deployed YouTube ranking system and targets long-term satisfaction.	Uses real YouTube behavioural traffic and changes ranking. This study instead uses primary user survey data and regression to identify factors associated with recommendation reliance.
Dokoupil et al. (2024)	Controlled user study with recommendation lists	Comparison of diversity metrics and diversification thresholds	User responses / perceived diversity	Users preferred metadata/content-based diversity; perception was not linear for small diversity differences.	Both studies use user-level preference information. This study uses broader YouTube behaviour and preference questions rather than controlled recommendation lists.
Coppolillo et al. (2024)	Five public recommendation datasets	User-behaviour model with relevance-diversity framework and copula-based recommendation strategy	Multiple benchmark datasets and comparison with recommendation baselines	Models user stopping behaviour and balances relevance with diversity.	Focuses on recommendation algorithm design. This study does not build a recommender; it analyses user-reported relevance, repetition, discovery and reliance.

5.1 Position of the Present Study

The three studies address recommendation systems at different levels. Wu et al. focus on production ranking and long-term satisfaction on YouTube. Dokoupil et al. focus on how users perceive diversity. Coppolillo et al. model the interaction between relevance, diversity and user stopping behaviour.

The present study is narrower in technical scope but directly combines several user-reported dimensions in one business-analytics dataset: recommendation reliance, viewing time, skip friction, repetition, surface usage, discovery, satisfaction, click and ignore motivations, and preferred recommendation strategies.

6. Results, Business Insights and Recommendations

6.1 Business Insights

The first finding is that recommendation reliance varies with usage intensity. Multiple-times-a-day users reported an average recommendation watch percentage of 52.85%, compared with 25.91% for users who use YouTube several times a week. Weekly hours also had a positive correlation with recommendation watch percentage ($r = +0.110$).

The second finding is that recommendation friction is associated with lower reliance. Repetition frequency had a negative correlation with recommendation watch percentage ($r = -0.057$). The daily-user analysis also showed a median decline from 50% to 40% as reported repetition moved from Rarely to Very Often.

The third finding is that surface breadth is positively associated with recommendation reliance. The number of surfaces used had the strongest positive simple correlation with the target ($r = +0.152$), and Omni_Surface users had a mean recommendation watch percentage of 53.6%, compared with 22.9% for Search_Oriented users.

The fourth finding is that severe skip friction can coincide with leaving the platform. Users who left YouTube after skipping more than 20 videos had a mean recommendation watch percentage of 22.9%. By contrast, users who continued browsing other recommendations had the highest mean reliance at 55.3%.

6.2 Recommendations Linked to Findings

- Reduce repetitive recommendation sequences. This follows the negative relationship between repetition frequency and recommendation watch percentage and the lower median reliance observed among daily users reporting Very Often repetition.
- Improve relevance before increasing recommendation volume. Irrelevant videos were the most common reported issue at 33.0%, and high skip friction was associated with lower reliance. Ranking should therefore place greater emphasis on matching the user's current interest before expanding the number of suggestions.
- Monitor high-skip sessions as a friction signal. The group that skipped more than 20 videos and then left YouTube had only 22.9% mean recommendation reliance. A large number of consecutive skips can therefore be treated as a signal to change the recommendation set or offer more direct search/control options.
- Maintain cross-surface consistency while supporting multiple discovery paths. Omni_Surface users had the highest mean reliance at 53.6%, while Search_Oriented users had 22.9%. Recommendations can be made more consistent across Home, Shorts and Subscription surfaces so that users who use several surfaces receive related but non-repetitive discovery opportunities.
- Use satisfaction and intended engagement as monitoring variables. Satisfaction showed a positive correlation with recommendation watch percentage ($r = +0.078$), while watch_more_likelihood was the most important Random Forest predictor (0.108). These variables can therefore be monitored alongside

interaction metrics rather than relying only on clicks. The complete model-importance and correlation outputs are provided in the accompanying analysis notebook.

6.3 Limitations

- The primary empirical sample contains 115 real responses; the 10,000-record dataset is synthetic and should not be described as 10,000 independent survey responses.
- The survey measures self-reported behaviour rather than platform event logs.
- The correlations are small for several variables and do not establish causality.
- The predictive models were evaluated on synthetic data derived from the same 115-response seed, so the reported performance may not generalise to a new real-user population.
- The project studies YouTube as a case study and does not have access to YouTube's internal recommendation features or ranking signals.

7. Conclusion

This case study examined recommendation-system interaction from a user perspective using YouTube as the case study. The primary survey of 115 respondents captured viewing intensity, recommendation reliance, click and ignore reasons, recommendation surfaces, skip behaviour, repetition, discovery, satisfaction and preferred recommendation strategies. The dataset was expanded to 10,000 synthetic records using an empirical Gaussian-copula resampling procedure for the predictive analysis. The detailed preprocessing, feature-engineering, validation and model outputs are provided in the accompanying analysis notebook.

The analysis found that surface breadth, weekly viewing hours and satisfaction had positive relationships with recommendation reliance, while skip friction and perceived repetition had negative relationships. Random Forest regression provided substantially lower prediction error than the Ridge baseline on the synthetic dataset. The business recommendations therefore focus on reducing repetition and high-skip sessions, improving relevance, and using satisfaction and engagement-intention signals alongside interaction measures.

8. References

1. Wu, Y., Chang, D., She, J., Zhao, Z., Wei, L., & Heldt, L. (2024). Learned Ranking Function: From Short-term Behavior Predictions to Long-term User Satisfaction. Proceedings of the ACM Conference on Recommender Systems (RecSys 2024). <https://arxiv.org/abs/2408.06512>
2. Dokoupil, P., Boratto, L., & Peska, L. (2024). User Perceptions of Diversity in Recommender Systems. Proceedings of the 32nd ACM Conference on User Modeling, Adaptation and Personalization (UMAP 2024), 212–222. <https://doi.org/10.1145/3627043.3659555>
3. Coppolillo, E., Manco, G., & Gionis, A. (2024). Relevance Meets Diversity: A User-Centric Framework for Knowledge Exploration Through Recommendations. Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2024), 490–501. <https://doi.org/10.1145/3637528.3671949>